MODULE 11

Scan Matching: Fitting Clouds Together

11.1 The puzzle-piece idea

SIMPLE APPROACH

Draw a shape on a see-through sheet. Your friend draws the same shape on another sheet, but a little shifted and turned. Lay the sheets on top of each other and slide and twist the top one until the drawings line up perfectly. How far you slid and twisted it tells you how your friend's sheet was placed! A LiDAR does this with two scans: sliding and twisting the new scan until it lines up with the old one tells it how far the car moved.

ENGINEER'S VIEW

Scan matching (registration) finds the rigid transform $(\mathbf{R}, \mathbf{t})$ that best aligns a source cloud $\{\mathbf{p}_i\}$ to a target cloud or map $\{\mathbf{q}_i\}$. Matching each new scan to the previous one gives *scan-to-scan* odometry; matching to an accumulated local map gives *scan-to-map* odometry, which drifts less. LIO-SAM matches each new keyframe to a local map built from its recent "sub-keyframes" (Module 12). The answer becomes a measurement (a factor) of how the vehicle moved.

11.2 Iterative Closest Point (ICP)

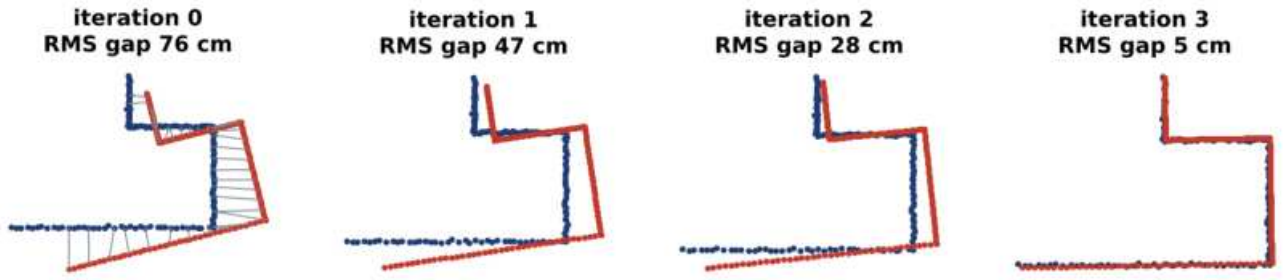


Figure 11.1: A real ICP run: pair each red point with its nearest blue point, solve the best fit, move, repeat.

KEY EQUATION — The ICP objective

$$\min_{\mathbf{R},\mathbf{t}} \; E = \sum_{i=1}^{N} \left\| \mathbf{R}\mathbf{p}_i + \mathbf{t} - \mathbf{q}_{c(i)} \right\|^2$$

What every symbol means

Symbol	Meaning	Units
$\mathbf{p}_i$	point i of the new (source) scan	m
$\mathbf{q}_{c(i)}$	its corresponding (closest) point in the target	m
$c(i)$	the correspondence: which target point is paired with $\mathbf{p}_i$	—
$\mathbf{R}, \mathbf{t}$	rotation and translation being solved for	—, m
$\|\cdot\|^2$	squared distance	m^2
E	total squared misfit (least-squares cost)	m^2

The algorithm alternates two easy problems:

1. **Associate:** for the current guess, pair every source point with its nearest target point (a k-d tree makes this fast).
2. **Solve:** with pairs fixed, find the best $(\mathbf{R}, \mathbf{t})$ in closed form (below).
3. **Apply** the transform and **repeat** until the cost stops decreasing.

ICP converges to the nearest local minimum, so it needs a good starting guess — which the IMU provides.

KEY EQUATION — Closed-form best fit (Arun/Horn/Umeyama)

$$\bar{\mathbf{p}} = \tfrac{1}{N}\textstyle\sum \mathbf{p}_i, \quad \bar{\mathbf{q}} = \tfrac{1}{N}\textstyle\sum \mathbf{q}_i, \quad \mathbf{W} = \textstyle\sum (\mathbf{p}_i - \bar{\mathbf{p}})(\mathbf{q}_i - \bar{\mathbf{q}})^{\mathsf{T}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathsf{T}}, \quad \mathbf{R} = \mathbf{V}\mathbf{U}^{\mathsf{T}}, \quad \mathbf{t} = \bar{\mathbf{q}} - \mathbf{R}\bar{\mathbf{p}}$$

In 2-D this reduces to $\theta = \operatorname{atan2}\left(\textstyle\sum (p'_x q'_y - p'_y q'_x),\ \sum (p'_x q'_x + p'_y q'_y) \right)$.

What every symbol means

Symbol	Meaning	Units
$\bar{\mathbf{p}}, \bar{\mathbf{q}}$	centroids (average points) of the two sets	m
$\mathbf{p}', \mathbf{q}'$	points with their centroid subtracted ("centered")	m
$\mathbf{W}$	cross-covariance matrix of the centered sets	m^2
$\mathbf{U}, \mathbf{\Sigma}, \mathbf{V}$	singular value decomposition of $\mathbf{W}$; if $\det(\mathbf{V}\mathbf{U}^{\mathsf{T}}) = -1$ flip the last column of $\mathbf{V}$ to avoid a mirror image	—

WORKED EXAMPLE — best fit of four matched points

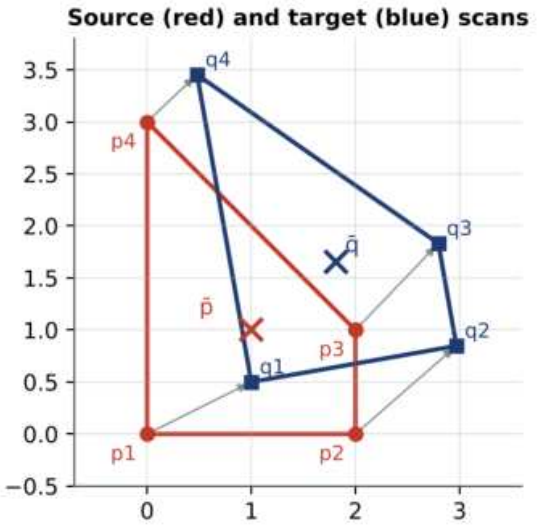
Given: source $\mathbf{p}_{1..4} = (0,0), (2,0), (2,1), (0,3)$; target $\mathbf{q}_{1..4} = (1.000, 0.500), (2.970, 0.847), (2.796, 1.832), (0.479, 3.454)$, already paired.

Step 1 — centroids

$\bar{\mathbf{p}} = (1,1)$; $\bar{\mathbf{q}} = (1.811, 1.658)$

Step 2 — center and sum

(table below)



p'_x	p'_y	q'_x	q'_y	$p'_x q'_x + p'_y q'_y$	$p'_x q'_y - p'_y q'_x$
−1	−1	−0.811	−1.158	1.970	0.347
1	−1	1.158	−0.811	1.970	0.347
1	0	0.985	0.174	0.985	0.174
−1	2	−1.332	1.796	4.924	0.868
			Sum	**9.848**	**1.736**

WORKED EXAMPLE — (continued)

Step 3 — rotation

$\theta = \text{atan2}(1.736,\ 9.848) = 10.00°$

Step 4 — translation

$\mathbf{t} = \bar{\mathbf{q}} - \mathbf{R}\bar{\mathbf{p}}$: $\mathbf{R}\bar{\mathbf{p}} = (0.9848 - 0.1736,\ 0.1736 + 0.9848) = (0.811,\ 1.158)$; $\mathbf{t} = (1.811 - 0.811,\ 1.658 - 1.158) = (1.000,\ 0.500)$

Step 5 — check with SVD

$\mathbf{W} = \begin{pmatrix} 4.287 & -1.275 \\ -3.012 & 5.562 \end{pmatrix}$ gives $\mathbf{VU}^{\mathsf{T}} = \begin{pmatrix} 0.985 & -0.174 \\ 0.174 & 0.985 \end{pmatrix} = \mathbf{R}(10°)$ ✓

Result: the second scan is the first rotated 10° and shifted (1.0, 0.5) m — exactly how the target was made. In real ICP the pairs are only guesses, so steps 1–5 repeat.

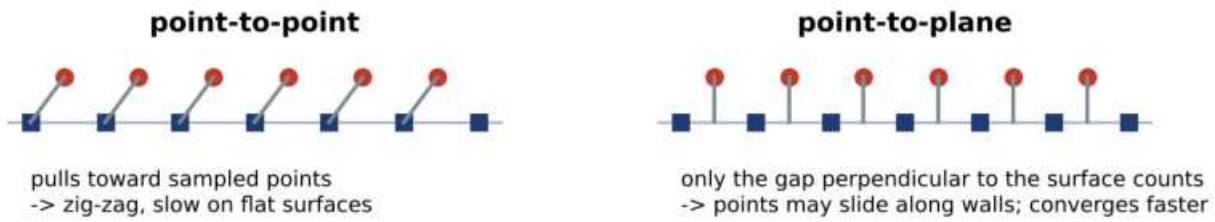


Figure 11.2: Point-to-point pulls toward sample points; point-to-plane only penalizes the gap normal to the surface.

11.3 Better cost functions: point-to-plane and NDT

KEY EQUATION — Point-to-plane ICP

$$E = \sum_i \Big((\mathbf{R}\mathbf{p}_i + \mathbf{t} - \mathbf{q}_i) \cdot \mathbf{n}_i \Big)^2$$

What every symbol means

Symbol	Meaning	Units
$\mathbf{n}_i$	unit normal of the target surface at $\mathbf{q}_i$	—
$(\cdot) \cdot \mathbf{n}_i$	dot product: keeps only the part of the gap perpendicular to the surface	m

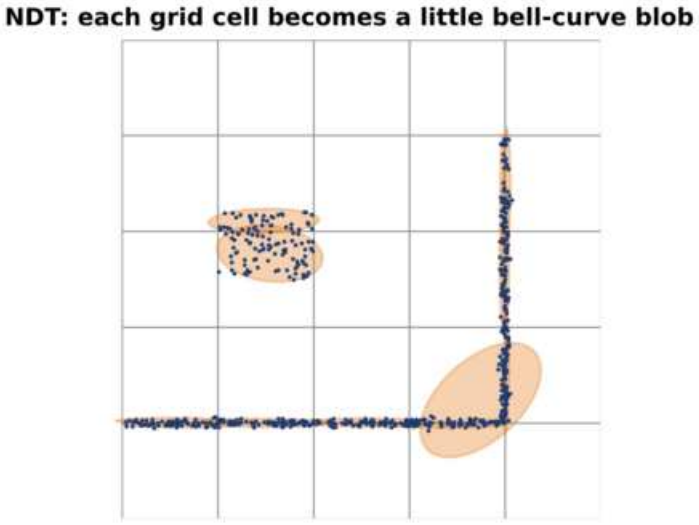


Figure 11.3: The Normal Distributions Transform summarizes each grid cell by a mean and covariance.

ENGINEER'S VIEW

NDT (Normal Distributions Transform) divides the target into voxels and fits each with a Gaussian (mean $\boldsymbol{\mu}_k$, covariance $\boldsymbol{\Sigma}_k$). The source is scored by $\sum_i \exp\big(-\frac{1}{2}(\mathbf{x}_i - \boldsymbol{\mu}_k)^\mathsf{T}\boldsymbol{\Sigma}_k^{-1}(\mathbf{x}_i - \boldsymbol{\mu}_k)\big)$ with $\mathbf{x}_i = \mathbf{R}\mathbf{p}_i + \mathbf{t}$, and maximized. No nearest-neighbor search is needed, it is smooth, and it is widely used in automotive localization. **Generalized ICP** (GICP) blends both ideas by modeling local surface covariances on both clouds.

11.4 Feature-based matching (LOAM and LIO-SAM)

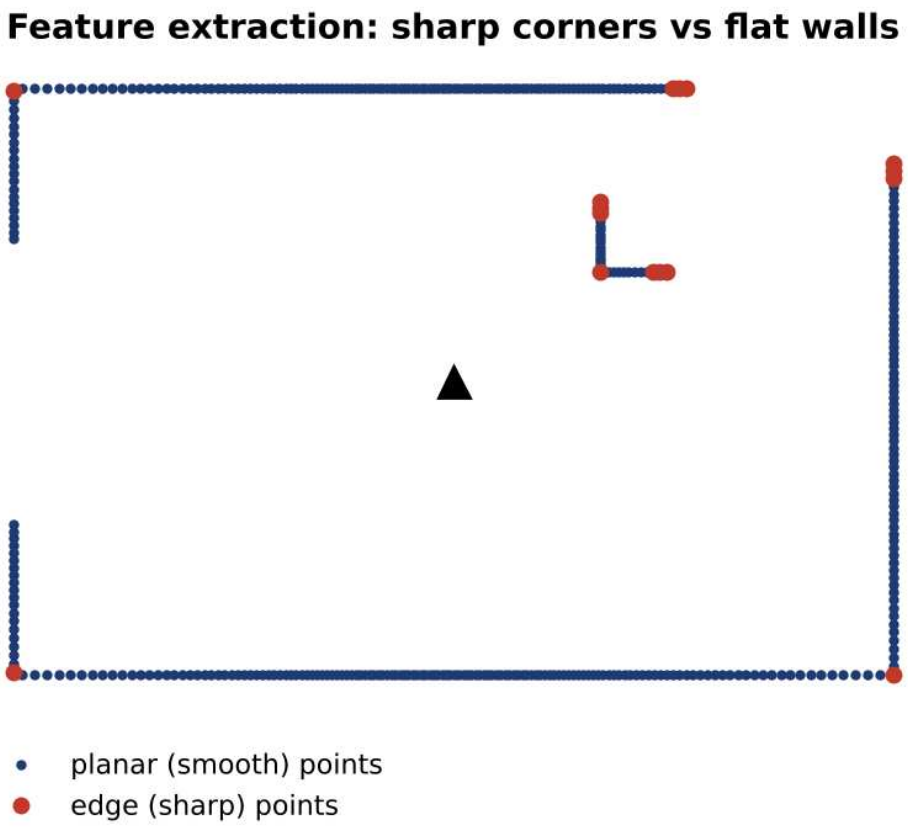


Figure 11.4: Points on sharp corners become edge features; points on smooth surfaces become planar features.

SIMPLE APPROACH

Instead of matching every single dot, look for the *special* ones: sharp corners (like the corner of a box) and flat sides (like a table top). Corners are easy to recognize and flat sides are easy to slide against. Matching only these is much faster.

KEY EQUATION — Roughness (curvature) score

$$c_i = \frac{1}{|S| \cdot \rho_i} \left| \sum_{j \in S} (\rho_j - \rho_i) \right|$$

What every symbol means

Symbol	Meaning	Units
ρ_i	range of point i along one scan line	m
S	its neighbors on the same scan line ($\lvert S \rvert$ = how many, e.g. 5 each side)	—
c_i	roughness: large ⇒ edge feature; small ⇒ planar feature	—

(Simplified range form; LOAM uses the same expression with 3-D point vectors.)

WORKED EXAMPLE — edge or plane?

Given: two neighbors each side ($|S| = 4$).

Flat wall

ranges 10.00, 10.01, 10.00, 9.99, 10.00: sum of differences $= 0 + 0.01 - 0.01 + 0 = 0 \Rightarrow c = 0$ **planar**

Corner

ranges 10.0, 10.0, 8.0, 8.0, 8.0: sum $= 2 + 2 + 0 + 0 = 4 \Rightarrow c = 4/(4 \times 8) = 0.125$ **edge**

Result: thresholds (e.g. $c > 0.1$ edge, $c < 0.1$ plane in LOAM's default) sort the points.

11.4.1 Distances to edges and planes (LIO-SAM eqs. 10–11)

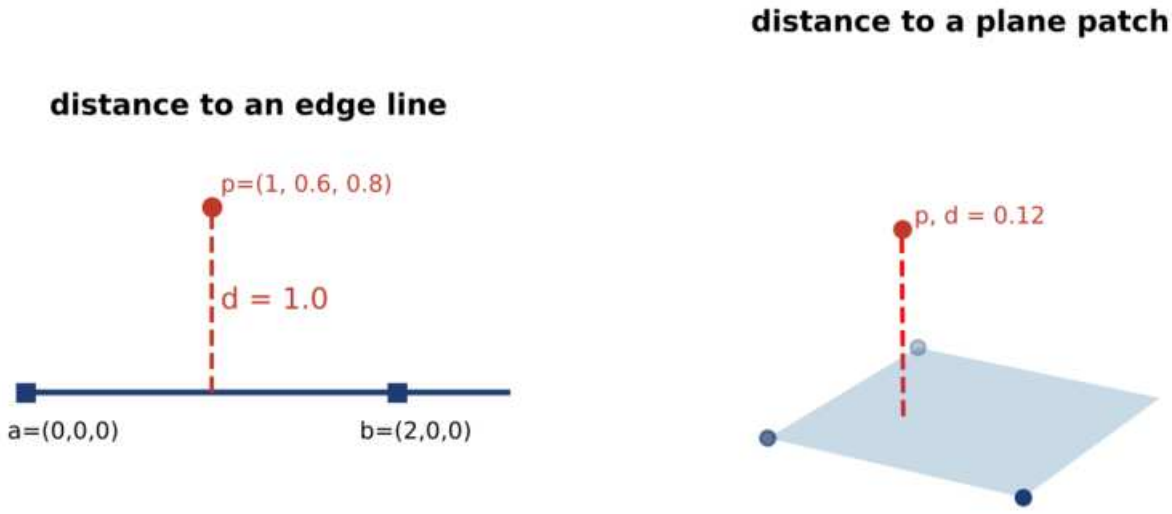


Figure 11.5: An edge feature is compared with a line through two map points; a planar feature with a plane through three.

KEY EQUATION — Point-to-line and point-to-plane distances

$$d_e = \frac{\left\|(\mathbf{p}-\mathbf{a}) \times (\mathbf{p}-\mathbf{b})\right\|}{\|\mathbf{a}-\mathbf{b}\|} \qquad d_p = \frac{\left|(\mathbf{p}-\mathbf{u}) \cdot \left((\mathbf{u}-\mathbf{v}) \times (\mathbf{u}-\mathbf{w})\right)\right|}{\left\|(\mathbf{u}-\mathbf{v}) \times (\mathbf{u}-\mathbf{w})\right\|}$$

What every symbol means

Symbol	Meaning	Units
$\mathbf{p}$	a feature point of the new scan, transformed into the map frame with the current guess	m
$\mathbf{a}, \mathbf{b}$	two map points defining the matching edge line ($\mathbf{p}^e_{i,u}, \mathbf{p}^e_{i,v}$ in the paper)	m
$\mathbf{u}, \mathbf{v}, \mathbf{w}$	three map points defining the matching planar patch	m
$\times$	cross product (Module 2): its length is a parallelogram area; area ÷ base = height = distance	—
$(\mathbf{u}-\mathbf{v}) \times (\mathbf{u}-\mathbf{w})$	a vector normal to the plane	m^2

WORKED EXAMPLE — edge and plane distances

Edge

$\mathbf{a} = (0,0,0)$, $\mathbf{b} = (2,0,0)$, $\mathbf{p} = (1,0.6,0.8)$. $\mathbf{p}-\mathbf{a} = (1,0.6,0.8)$, $\mathbf{p}-\mathbf{b} = (-1,0.6,0.8)$; cross $= (0.6\cdot 0.8 - 0.8\cdot 0.6,\ 0.8\cdot(-1) - 1\cdot 0.8,\ 1\cdot 0.6 - 0.6\cdot(-1)) = (0,-1.6,1.2)$; length $= \sqrt{2.56+1.44} = 2.0$; divide by $\|\mathbf{a}-\mathbf{b}\| = 2$: $d_e = 1.0$ m.

Plane

$\mathbf{u} = (0,0,0)$, $\mathbf{v} = (1,0,0)$, $\mathbf{w} = (0,1,0)$, $\mathbf{p} = (0.3,0.4,0.12)$. Normal $= (\mathbf{u}-\mathbf{v}) \times (\mathbf{u}-\mathbf{w}) = (-1,0,0)\times(0,-1,0) = (0,0,1)$; $d_p = |(0.3,0.4,0.12)\cdot(0,0,1)|/1 = 0.12$ m.

Result: LIO-SAM then finds the pose that minimizes $\sum d_e + \sum d_p$ over all features using the Gauss–Newton method.

ENGINEER'S VIEW

Gauss–Newton in one paragraph. Stack all distances into a residual vector $\mathbf{d}(\mathbf{x})$ that depends on the 6-DOF pose $\mathbf{x}$. Linearize: $\mathbf{d}(\mathbf{x}+\delta\mathbf{x}) \approx \mathbf{d} + \mathbf{J}\delta\mathbf{x}$, where $\mathbf{J}$ is the Jacobian. Setting the derivative of $\|\mathbf{d}+\mathbf{J}\delta\mathbf{x}\|^2$ to zero gives the *normal equations* $\mathbf{J}^\mathsf{T}\mathbf{J}\,\delta\mathbf{x} = -\mathbf{J}^\mathsf{T}\mathbf{d}$ — exactly the normal equations of a survey least-squares adjustment. Solve, update $\mathbf{x} \leftarrow \mathbf{x}+\delta\mathbf{x}$, repeat until $\delta\mathbf{x}$ is tiny.

11.5 When matching fails: the hallway effect

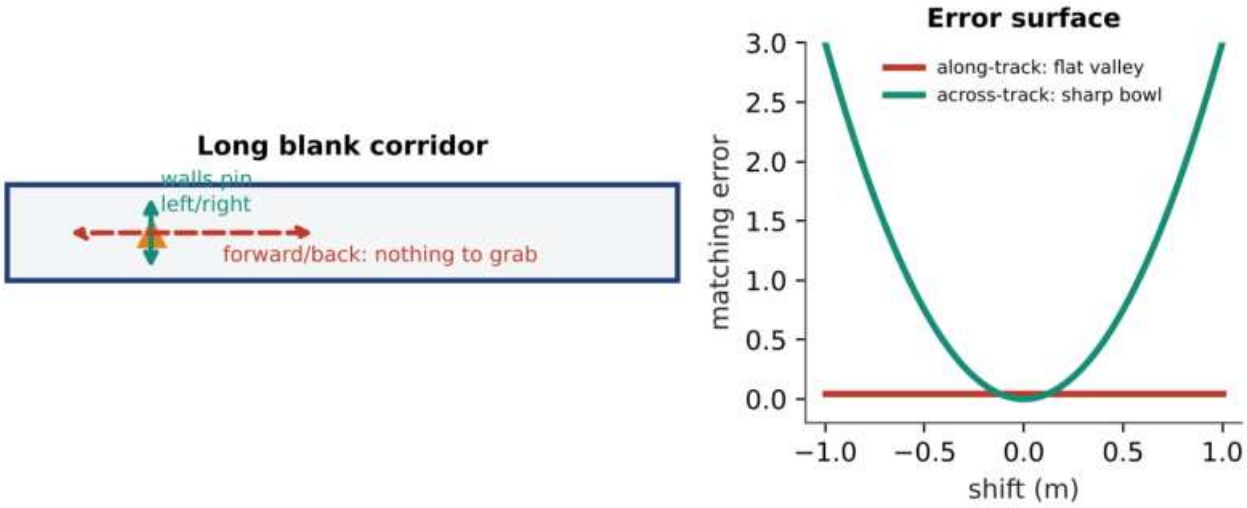


Figure 11.6: In a featureless corridor the matching error has a flat valley along the corridor — any forward shift fits equally well.

ENGINEER'S VIEW

Scan matching can only measure motion in directions where geometry changes. In a long uniform corridor, tunnel, or a highway with continuous barriers, the cost is flat along the axis, $\mathbf{J}^\mathsf{T}\mathbf{J}$ is nearly singular in that direction, and the estimate is free to stretch or shrink. Mathematically, the smallest eigenvalue of $\mathbf{J}^\mathsf{T}\mathbf{J}$ is near zero; some systems monitor this "degeneracy" and down-weight LiDAR along that axis. Surveyors recognize this as a traverse with poor geometry in one direction.

SURVEYOR'S FIELD TIP

In corridors: add control or targets along the length, walk/drive both directions, include side rooms or cross-ties to create unique geometry, place temporary objects (cones, boxes) if needed, and always check along-track distances against a taped or total-station measurement.

TRY IT YOURSELF

11.1 Source points $(0,0),(1,0)$ match target points $(2,1),(2,2)$. Find θ and $\mathbf{t}$ by the closed form.
11.2 Distance from $\mathbf{p} = (3,4,5)$ to the line through $(0,0,0)$ and $(0,0,10)$?
11.3 (Simple) Why is it hard to tell how far you have walked in a long, plain hallway with your eyes half-closed?

Answers: worked solutions are in Appendix A, "Answers to Try It Yourself," of the Reference Book, starting on page 158.

SOURCES FOR THIS MODULE

Besl & McKay (1992), *IEEE PAMI* 14(2); Arun, Huang & Blostein (1987), *IEEE PAMI* 9(5); Segal *et al.* (2009), Generalized-ICP; Biber & Straßer (2003), NDT; Zhang & Singh (2017), LOAM, *Autonomous Robots* 41(2); Shan *et al.* (2020), eqs. (10)–(11).